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28 November 1980
Superseding
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11 May 1960

MILITARY SPECIFICATION

MODULATOR, DATA: MD-242/MS

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of data modulator, hereinafter referred to as "modulator". The modulator delivers 400 cycles per second and is used with antiaircraft and guided missile fire control systems (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATION

Military

MIL-F-13926	Fire Control Materiel; General Specification Governing the Manufacture and Inspection of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of US Military Property

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document.

FSC 1285

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MIL-STD-202	Commercial Packaging of Supplies and Equipment
MIL-STD-1188	Test Methods for Electronic and Electrical Component parts

DRAWINGS

Army

C7616083	Oscillator
E7620638	Modulator, Data: MD-242/MS
F7620639	Modulator, Data: MD-242/MS Schematic Diagram

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Production model. At the beginning of production, one modulator, representative of production, shall be made available to the Government to determine if the modulator meets, all requirements of this specification.

3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction. The modulator shall be manufactured in accordance with drawing E7620638, "Modulator, Data: MD-242/MS" and drawings pertaining thereto. The requirements of this specification are detailed only to extent considered necessary to obtain the desired mechanical and electrical characteristics, performance and permanence capabilities prior to final acceptance by the Government.

3.4 Ground potential. The design and construction of the modulator shall insure that all external parts are at ground potential at all times.

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3.5 General specification. The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

a. Order of precedence
b. Protective finished surfaces
c. Special accuracy
d. Finish, painting and protective
e. Electron tubes
f. Workmanship

3.6 Environmental

3.6.1 Vibration. The modulator shall be subjected to a simple harmonic motion having an amplitude of 0.03-inch (0.06-inch maximum excursion). The frequency shall be varied uniformly between the limits of 10 and 55 cycles per second return to 10 cycles and shall be traversed in approximately 1 minute. This motion shall be applied for a period of 10 minutes in each of the 3 mutually perpendicular directions (total of 30 minutes). Subsequent to vibration the modulator shall not be damaged or adversely affected and shall meet all requirements of this specification.

3.6.2 Operating temperatures. The modulator shall meet all the performance requirements of this specification when thermally stabilized at ambient temperatures of minus 40 degrees and plus 150 degrees Fahrenheit.

3.6.3 Storage temperatures. The modulator shall show no evidence of physical failure and shall meet all performance requirements of this specification at room temperature (plus 60 to 90 degrees Fahrenheit) after having been thermally stabilized at ambient temperatures of plus 160 degrees and minus 80 degrees Fahrenheit.

3.7 Electrical.

3.7.1 Circuitry. The circuitry of the modulator shall be in accordance with Electrical Schematic, drawing F7620639.

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3.7.2 Dielectric strength. There shall be no puncture of insulation, no sparkover, or decrease in dielectric strength, when the insulated areas of the modulator are subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than 5 seconds.

3.7.3 Insulation resistance. Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

3.8 Performance. When set-up in accordance with pretest information specified in Section 4, the modulator shall meet the following performance requirements. All symbols referred to are those shown on drawing F7620639.

3.8.1 Test oscillator level. The output control of the test oscillator shall be capable of adjusting for a reading of 24 plus and minus 0.25 volts when switches S8 and S9 are on position 1.

3.8.2 Capacitor, C5

3.8.2.1 The voltmeter shall read between 0.35 to 0.46 volts, when the following switches are set on their respective positions:

SWITCHE	POSITION
S1	A
S2	1
S4	1
S5	1
S8	1
S9	2
S10	on
S11	on

3.8.2.2 The voltmeter shall read between 0.35 to 0.46 volt, when S1 is switched to position B.

3.8.3 Output network

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"3.8.3.1 The voltmeter shall read between 1.35 and 1.65 volts, when the specified switches are set to their indicated positions:

Switch	Position	Switch	Position
S1	A	S7	1
S2	2	S8	1
		S9	3"

3.8.3.2 The voltmeter shall read between 1.35 to 1.65 volts, when S1 is switched to position B.

3.8.4 Balance and noise

3.8.4.1 The modulator balance control R23 shall be capable of adjusting for a 0.012 volt (max.) reading on the voltmeter, when the following switches are set on their respective positions:

SWITCHES	POSITION
S1	A
S2	2
S4	1
S5	4
S7	2
S8	2
S9	3

3.8.4.2 The voltmeter reading shall not increase to more than 0.02 volt, when S6 is pressed and held for 20 to 30 seconds and all switches retained as in 3.8.4.1.

3.8.4.3 Modulator balance control R53 shall be capable of adjusting for a .012 volt (max.) reading on the voltmeter, when S1 is switched to position B.

3.8.4.4 The voltmeter reading shall not increase to more than 0.02 volt, when S6 is pressed and held for 20 to 30 seconds.

3.8.5 Gain

3.8.5.1 The voltmeter shall read between 0.015 to 0.23 volt, when the following switches are set on their respective positions:

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SWITCH	POSITION
S1	A
S2	2
S3	+250 V
S4	2
S5	1
S7	3
S8	2
S9	3

3.8.5.2 The voltmeter shall read between 0.015 to 0.023 volt, when S1 is switched to position B.

3.8.5.3 The voltmeter shall read between 0.015 to 0.023 volt, when S1 is returned to position A and S3 is switched to the minus 250V position.

3.8.5.4 The voltmeter shall read between 0.015 to 0.023 volt, when S1 is switched to position B and S3 retained on the minus 250V position.

3.8.6 Overload

3.8.6.1 The voltmeter shall read between 0.11 to 0.13 volt for either position of S3, when the following switches are set on their respective positions:

SWITCH	POSITION
S1	A
S4	3
S5	1
S7	3
S8	2
S9	3

3.8.6.2 The voltmeter shall read between 0.11 to 0.13 volt for either position of S3, when S1 is switched to position B and all other switches retained on the positions specified in 3.8.6.1 above.

3.8.7 Phase

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3.8.7.1 The vertical and horizontal gain of the oscilloscope shall be capable of adjustment to give an elliptical pattern inclined at approximately 45 degrees, when the following switches are set in their respective positions.

SWITCH	POSITION
S1	A
S3	250V
S4	3
S5	1
S7	3
S8	3
S9	3

3.8.7.2 Resistor, R2, of the test panel, shall be capable of narrowing the ellipse to a single line, when S1 is on either position A or B.

3.8.8 Input network

3.8.8.1 Resistor, R1, of the test panel, shall be capable of adjustment to have the voltmeter read 0.059 to 0.061 volt, when the following switches are set in their respective positions:

SWITCH	POSITION
S1	A
S2	3
S4	1
S5	1
S7	3
S8	1
S9	3

3.8.8.2 The voltmeter shall read between 0.054 to 0.066 volt, when S2 is switched to position 4, S5 to position 2 and all other switches retained as specified in 3.8.8.1 above.

3.8.8.3 The voltmeter shall read between 0.059 to 0.061 volt, when switch S1 is switched to position B, switch 2 to 3, switch 5 to 1, and all other switches retained as specified in 3.8.8.1.

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3.8.8.4 The voltmeter shall read between 0.054 to 0.066 volt, when switch 2 is switched to position 4, switch S5 to 2, and all other switches retained as specified in 3.8.8.1.

3.9 Identification marking. Unless otherwise specified, identification marking shall be in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Contractor's quality assurance system. The contractor shall provide and maintain an adequate quality assurance system acceptable to the Government covering the supplies under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptance when it provides:

a. The quality assurance required by the applicable drawings and specifications listed in the contract or order.

b. Contractor's quality standards of acceptability.

c. Procedure for continually evaluating product quality.

d. Procedure for continually reporting evidence of actual product quality.

e. Measures for preventing any significant departure of product quality from standards.

4.2.1 The contractor will not be restricted to the inspection station or to the method of inspection listed provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure required by this specification shall be used. The contractor shall notify the Government of and obtain approval for any change to the written procedure that might affect the degree of assurance required by this specification and other applicable documents referenced herein.

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4.3 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine that the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have a detrimental affect upon the quality of the product, will be immediately called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered will be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.4 Production sample. Unless otherwise specified in the contract, one modulator shall be selected by the Government inspector as soon as practicable after the start of production, and be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.4.1 Production sample failure. Should the production sample fail to pass the prescribed examination and tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.5 Inspection provisions.

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4.5.1 Lot formation. Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with MIL-STD-105.

4.5.2 Inspection of components. Contractor's inspection for all components shall be in accordance with the inspection procedures of the written inspection plan required by 4.2 and 4.2.1. The plan shall include procedures for checking all positional and functional characteristics shown on the drawings which indicate precision requirements that have a bearing on the functional accuracy and performance of the unit.

4.5.2.1 Sampling inspection. Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table I.

TABLE I - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.5.2.2 Disposition of non-conforming product. Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall consititute a defective unit. Disposition of non-conforming components shall be in accordance with MIL-STD-105.

4.6 Control of materials. Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identificable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.7 Acceptance tests.

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4.7.1 Classification of defects. The classification of defects in Table II and III shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual nonconforming items.

4.7.2 Disposition of rejected lots. Rejected lots shall be screened for all defective characteristics. After corrections have been made, and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in MIL-STD-105.

TABLE II - CLASSIFICATION OF DEFECTS

Use Inspection Level L5, and Sampling Plan Table IV-A of MIL-STD-105

MAJOR: AQL 4.0%	Req't Par.	QA Par.	Test Method
1. Physical failure (post vibration)	3.6.1	4.9.1.1	Tactile-Visual
2. Test oscillator level	3.8.1	4.9.3.1	Standard Measurement Equipment (SME)
3. Capacitor C5	3.8.2	4.9.3.2	SME
4. Output network	3.8.3	4.9.3.3	SME
5. Balance and noise	3.8.4	4.9.3.4	SME
6. Gain	3.8.5	4.9.3.5	SME
7. Overload	3.8.6	4.9.3.6	SME
8. Phase	3.8.7	4.9.3.7	SME
9. Input network	3.8.8	4.9.3.8	SME

NOTE: The above tests shall be performed while tehermally stabilized at ambient temperatures of plus 150 degrees and minus 40 degrees Fahrenheit, except the vibration test which will be at standard ambient temperature (plus 60 degrees to plus 90 degrees Fahrenheit).

TABLE III - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of MIL-STD-105

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MAJOR: AQL 1.5%	Req't Par.	QA Par.	Test Method
1. Circuitry	3.7.1	4.9.2.1	Standard Measurement Equipment (SME)
2. Dielectric strength	3.7.2	4.9.2.2	SME
3. Insulation resistance	3.7.3	4.9.2.3	SME
4. Test oscillator level	3.8.1	4.9.3.1	SME
5. Capacitor C5	3.8.2	4.9.3.2	SME
6. Output network	3.8.3	4.9.3.3	SME
7. Balance & noise	3.8.4	4.9.3.4	SME
8. Gain	3.8.5	4.9.3.5	SME
9. Overload	3.8.6	4.9.3.6	SME
10. Phase	3.8.7	4.9.3.7	SME
11. Input network	3.8.8	4.9.3.8	SME
12. Identification marking	3.9	4.9.4	Visual

NOTE: The above tests shall be performed at standard ambient tempeature (plus 60 degrees to plus 90 degrees Fahrenheit).

4.8 Inspection equipment.

4.8.1 Contractor provisions. Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have avilable, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.8.2 Accuracy requirments. Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where

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the ten percent accuracy imposes impractical or very difficult design which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.8.2.1 Test equipment accuracy requirements. In addition to the above accuracy requirements specified in 4.8.2, the following described in the text.

Item Number	Description
1.	Oscilloscope, Dumont Type 208 or equal.
2.	Voltmeter, Input impedance 0.5 megohms plus or minus 5% shunted by 30 micro-microfarads plus or minus 5%. Accuracy of plus or minus 2% on all scales.
3.	Oscillator, providing 120 volts, 10 milliamperes, 400 plus or minus 1 cycle per second, single phase.
4.	Power supply (plus 250 volts plus or minus 2.5 volts, 30 milliamperes, ripple not rreater than 0.02 volt peak).
5.	Power supply (minus 250 volts plus or minus 2.5 volts, 10 milliamperes, ripple not to exceed 0.02 volt peak).
6.	Power supply (6.3 volts plus or minus 0.3 volt, 3.5 amperes, 400 cycles per second plus or minus 20 cycles per second phase.
7.	Test panel.

4.8.2.2 Device to test for vibration requirements. The device shall consist of a vibrating machine which will produce a motion in each of the 3 mutually perpendicular planes at a uniformly varied frequency between the limits of 10 and 55 cycles per second and return to 10 cycles. It shall have an amplitude of 0.03 inch (0.06 inch maximum excursion) and shall be equipped with a medium capable of holding the unit securely to the vibrating machine.

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4.8.3 Government provisions. Test equipment listed in 4.8.2.1 will be used for Government product inspection (reference 4.3(b), and to settle disputes (reference 4.2.1).

4.9 Test methods and procedures.

4.9.1 Environmental.

4.9.1.1 Vibration. Each modulator in the sample lot shall be mounted in its normal operating position on a special testing device and vibrated in accordance with method 201 of MIL-STD-202. Subsequent to vibration, the modulator shall be subjected to a visual and tactile examination for physical damage. The modulator shall then be subjected to all the tests specified herein.

Reference Designation	
C1	CAPACITOR, fixed, mica, 300V dc, 100 mmfd ± 1%
J1, J2	CONNECTOR, Recap, 15 female contact
R1	RESISTOR, variable, composition, 2w, 0.1 meg ±10%
R2	RESISTOR, variable, wire wound, 8w, 0.1, meg ±10%
R3	RESISTOR, fixed composition, 1w, 0.1 meg ± 1%
R4, R9	RESISTOR, fixed, wire wound, 1w, 80,000 ohm ±1%
R5	RESISTOR, fixed, wire wound, 1w, 45,000 ohm ±1%
R6	RESISTOR, fixed, wire wound, 1/2w, 15,000 ohm ±1%
R7	RESISTOR, fixed, wire wound, 3/4w, 30,000 ohm ±1%
R8	RESISTOR, fixed, wire wound, 1w, 40,000 ohm ±1%
R10	RESISTOR, fixed, wire wound, 1w, 1000 ohm +1%
R11	RESISTOR, fixed, wire wound, 3/4w, 24,000 ohm +1%
R12	RESISTOR, fixed, composition, 1w, 1 meg ± 1%

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R13	RESISTOR, fixed, composition, 1w, 1 meg ±1%
R14	RESISTOR, fixed, wire wound, 1/4w, 3000 ohm ±1%
R15	RESISTOR, fixed, wire wound, 1/4w, 900 ohm ±1%
R16	RESISTOR, fixed, wire wound, 1/4w, 100 ohm ±1%
S1	SWITCH, rotary, 5 pole, 2 position, 2 section
S2	SWITCH, rotary, 1 pole, 5 position, 2 section
S3	SWITCH, toggle, SPDT, 250V ac or dc, 2 amp, 125V ac or dc 5 amp
S4	SWITCH, rotary, 1 pole, 3 position, 1 section
S5	SWITCH, rotary, 1 pole, 4 position, 1 section
S6	SWITCH, push, SPST, normally closed, 250V ac or dc 6 amp
S7	SWITCH, rotary, 3 pole, 3 position, 2 section
S8	SWITCH, rotary, 3 pole, 3 position, 1 section
S9	SWITCH, rotary, 2 pole, 3 position, 1 section
S10	SWITCH, toggle, DPST, 250V ac or dc 2 amp, 125V ac or dc 5 amp
SS11	SWITCH, toggle, SPST, 250V ac or dc 2 amp 125V ac or dc 5 amp
T1	TRANSFORMER, input, truns ratio 2.5 to 1 (Wnd 4-5-5 to 1-2-3) Wnd 1-2-3, 827 ohm; Wnd 4-5-6, 3050 ohm
J3	CONNECTOR, receptacle, 7 male contact

Component - Parts List for Figure 1
Figure 1A

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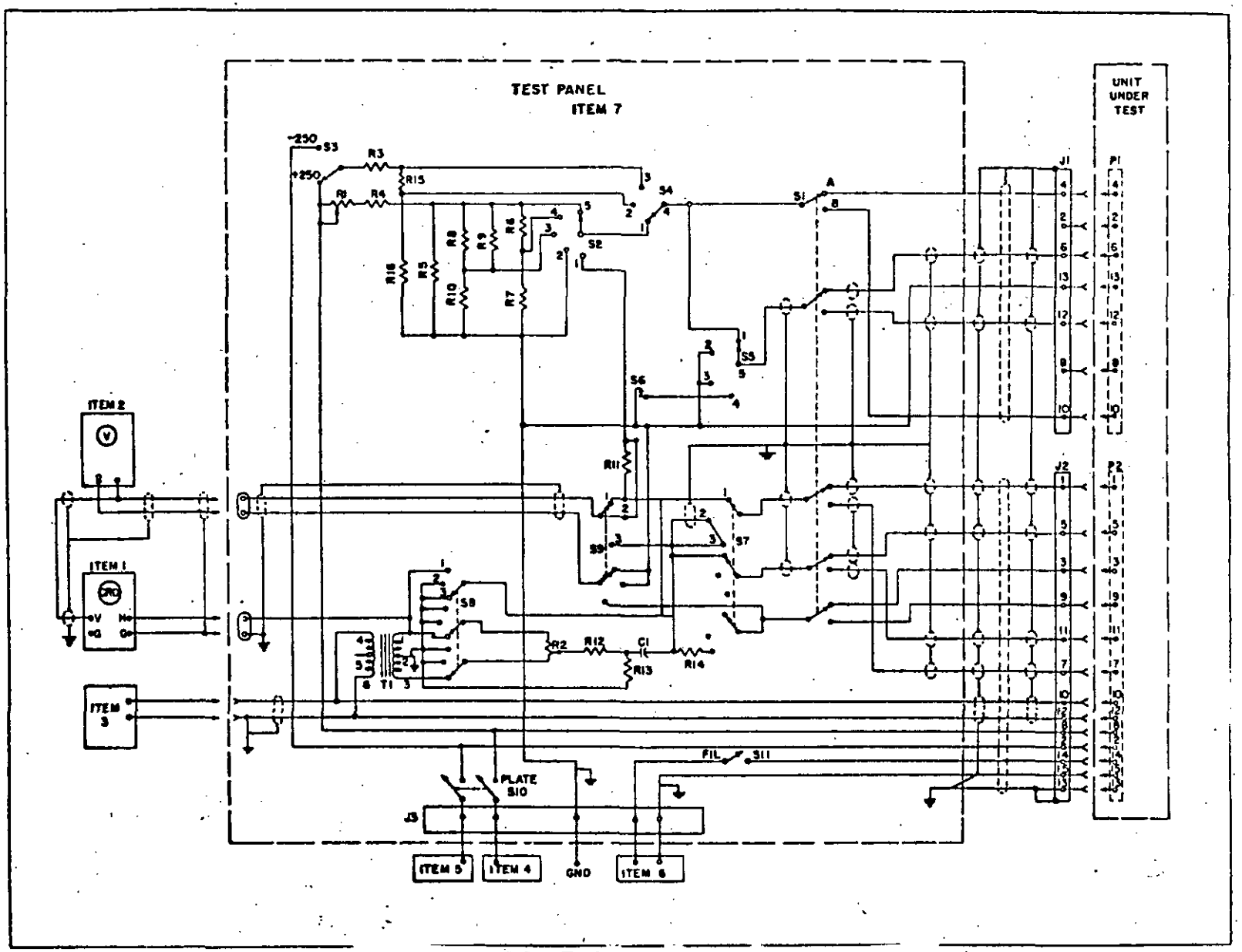


FIGURE 1A

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4.9.1.2 Operating temperature. This test shall be conducted in accordance with Procedure I of MIL-F-13926, except that the temperature shall not be reduced beyond condition "C" and not raised beyond condition "G". When the modulatur is thermally stabilized at condition "C" and "G", it shall be subjected to the tests specified in 4.9.3.1 to 4.9.3.8 inclusive and shall meet the requirements specified.

4.9.1.3 Storage temperature. This test shall be conducted in accordance with Procedure II of MIL-F-13926 to the conditions specified. Upon return to the standard ambient temperature, the modulator shall be inspected for damage and then subjected to the tests specified in 4.9.3.1 to 4.9.3.8 inclusive, and shall meet the requirements specified. This test shall be conducted on the production model only.

4.9.2 Electrical.

4.9.2.1 Circuitry. The modulator shall be tested, using standard measuring equipment, to demonstrate the continuity of the circuits in accordance with Electrical Schematic, drawing F7620639.

4.9.2.2 Dielectric strength. The modulator shall be tested in accordance with method 301 MIL-STD-202, and shall meet the requirements specified.

4.9.2.3 Insulation resistance. Immediately after satisfactorily withstanding the test for dielectric strength, the modulator shall be given the insulation resistance test in accordance with method 302, test condition "B", of MIL-STD-202, and shall meet the requirements specified.

4.9.3 Performance.

4.9.3.1 Test oscillator level. Switches S8 and S9 shall be set at position 1. The output control of the test oscillator, item 3, shall be capable of being adjusted to the specified voltage when read with AC voltmeter, item 2.

4.9.3.2 Capacitor C5 check.

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4.9.3.2.1 Set switches S1 at position A, S2 at 1, S4 ad 1, S5 at 1, S8 at 1, S9 at 2, S10 at on and S11 at on, Settings for all other switches are immaterial. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.2.2 Switch S1 shall be rotated to position "B". Settings for all other switches shall remain as in 4.9.3.2.1. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.3 Output network test.

4.9.3.1 Set switches S1 at position "A". S7 at 1, S8 at 1, S9 at 3. Settings for other switches are immaterial. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.3.2 Switch S1 shall be rotated to position "B". Settings for all other switches shall remain as in 4.9.3.3.1. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.4 Balance and noise test.

4.9.3.4.1 Set switch S1 at position "A", S2 at position "A", S2 at 2, S4 at 1, S5 at 4, S7 at 2, S8 at 2, and S9 at 3. Settings for other switches are immaterial. The modulator balance control R23 shall be capable of being adjusted to the specified voltage when read on voltmeter, item 2.

4.9.3.4.2 Press switch S6 the required time. The reading on voltmeter, item 2, shall not exceed the specified voltage.

4.9.3.4.3 Rotate switch S1 to position "B". The modulator balance control R53 shall be capable of being adjusted to the specified voltage when read on voltmeter, item 2.

4.9.3.4.4 Press switch S6 the required time. The reading on voltmeter, item 2, shall not exceed the specified voltage.

4.9.3.4.5 Potentiometers R23 and R53 on the modulator under test shall be locked.

4.9.3.5 Gain.

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4.9.3.5.1 Set switches S1 at position "A", S2 at 2, S3 at plus 250 volts, S4 at 2, S5 at 1, S7 at 3, S8 at 2, and S9 at 3. Voltmeter, item 2, shall read between the specified voltages.

4.9.3.5.2 Rotate switch S1 to position "B". Settings for all other switches shall remain as in 4.9.3.5.1. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.5.3 Return switch S1 to position "A" and rotate switche S3 to the minus 250 V position. Settings for all other switches shall remain as in 4.9.3.5. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.5.4 Rotate switch S1 to position "B" and S3 retained on the minus 250V position. Settings, for all other switches shall remain in 4.9.3.5.1. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.6 Overload.

4.9.3.6.1 Set switches S1 at position "A", S4 at 3, S5 at 1, S7 at 3, S8 at 2, and S9 at 3. Voltmeter, item 2, shall read between the voltages specified for either position of switch S3.

4.9.3.6.2 Set switch S1 to position "B" and repeat test procedure as specified in 4.9.3.6.1.

4.9.3.7 Phase.

4.9.3.7.1 Set switches S1 at position "A". S3 at plus 250V, S4 at 3, S5 at 1, S7 at 3, S8 at 3, and S9 at 3. Settings for all other switches are immaterial.

4.9.3.7.2 The signal shall be removed from the vertical input of the oscilloscope, item 1. The horizontal input shall be connected to the vertical input. The vertical gain and horizontal gain of the cathode ray tube shall be adjusted to give an inclined line. The connection between the horizontal and vertical inputs shall be removed and the signal reapplied to the vertical input.

4.9.3.7.3 The vertical and horizontal gain shall be adjusted to give an elliptical pattern inclined in the same direction as the line obtained in 4.9.3.7.1 at approximately the specified angle.

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4.9.3.7.4 Resistor R2, in Figure 1, of test panel, item 7 shall be adjusted until the ellips on oscilloscope, item 1, narrows to a single line, when S1 is on either position "A" or "B".

4.9.3.8 *Input network*.

4.9.3.8.1 Set switch S1 at position "A". S2 at 3, S4 at 1, S5 at 1, S7 at 3, S8 at 1, and S9 at 3.

4.9.3.8.2 Resistor R1 in Figure 1 of test panel, item 7, shall be ajusted to make voltmeter, item 2, read between the voltages specified.

4.9.3.8.3 Switch S2 shall be rotated to position 4, and S5 to position 2; all other switches shall remain as in 4.9.3.8.1. Voltmeter, item 2, shall read between the voltages specified.

4.9.3.8.4 Switch S1 shall be rotated to position "B", switch S2 to 3, switch S5 to 1, and other switches shall remain as in 4.9.3.8.1. Resistor R1 shall be adjusted to make voltmeter, item 2, read between the voltages specified.

4.9.3.8.5 Switch S2 shall be rotated to position 4, switch S5 to 2, and other switches shall remain as in 4.9.3.8.1. Voltmeter, item 2, shall read between the voltages specified.

4.9.4 Identification marking. Marking shall be visually inspected for conformance to MIL-STD-130.

4.9.5 Quality assurance provisions for packaging, packing and marking. Quality assurance for packaging, packing, and marking shall be inspected for conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation, packing, unitization, and marking - Level A, B or commercial packaging (see 6.2).

5.1.1 Level A or B. Preservation, packing, unitization, and marking shall be in accordance with packaging data sheet APN 7620638 and specifications referenced therein.

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5.1.2 Commercial. Preservation, packing, unitization and marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The modulator is intended to receive dc error signals from a dc, amplifier and deliver corresponding 400 cps error signals to one of twelve low power servo amplifiers. The amplified signals of the low power servo amplifiers drive the servo motor generators that position the pens of the horizontal-range, present-altitude, and predicted altitude plotting boards; and operate the firing-azimuth, firing-elevation, time-of-flight, and fuze servos. The modulator is a component of the following assemblies and systems:

Assembly	System
Computer 7604074	Antiaircraft Fire Control System M33C (90mm)
Computer 7603993	Antiaircraft Fire Control System M33D (120mm)
Amplifier-Relay Group OA-734/MSA-7	Radar Course Directing Central AN/MSW-1 and AN/MSW-3
Amplifier Computer Group OS-611/MSG-3	Guided Missile Fire Control System AN/MSG-3
Servo, Data Unit Test 80150950	Sets, Kits, and Outfits 8151370
Console, Upper (No. 3) 80150570	Sets, Kits, and Outfits 8151370
Panel, Test Electrical 8150303	Sets, Kits, and Outfits 8151370

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification

b. Selection of applicable level of preservation, packaging and packing.

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c. Applicable stock number.

d. Provisions for production model including date of delivery (see 3.1).

6.3 *Definitions*.

6.3.1 Standard measuring equipment (SME). Standard measuring equipment (SME) is defined as the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.2 Tactile. Skillful manipulation or sensing.

6.4 Design information. Design information regarding specific equipment not covered or referenced by this specification may be obtained from the procuring activity upon the specific request of the contractor.

Custodian:
Army-MI

Preparing activity:
Army-MI

Project No. 1285-A 527

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-45670A(MI) MODULATOR, DATA: MD-242/MS

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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